HUMBER POLYTECHNIC | Faculty of Applied Sciences & Technology

MENG 3050-Power Transmission Components

Assignment 03- Mixer Shaft Design **Worth: 10% of course grade**

Submission date: November 03,2026 Late

submission:

One day (-1mark)	Two days (-2marks)	Three days (-3 marks)	Four days (-4 marks)	Five days (-5 marks)

Introduction:

Industrial dough mixers use rotating shafts to transmit mechanical power from the motor and gearbox to the mixing hook, paddle, or blade, enabling the kneading and mixing of large quantities of dough. During operation, the shaft experiences combined torsional and bending loads resulting from the resistance of the dough and the forces acting on the mixing element. The shaft assembly must provide adequate strength, stiffness, and alignment while accommodating suitable bearing arrangements and a secure, removable connection for cleaning and maintenance.

In this assignment, students are required to design a mixing hook and drive shaft for a commercial dough mixer. The design must address material selection, shaft dimensions, bearing locations, stress concentrations, and the connection between the shaft and the mixing hook.

Students must apply machine design principles to determine appropriate dimensions, verify the shaft's strength under the expected operating loads, and produce a fully dimensioned engineering drawing of the proposed assembly.

Problem Details and Restrictions

The shaft assembly consists of a vertically oriented drive shaft supported by two bearings near its upper end, with a sprocket or gear and a mixing hook attachment at the lower end.

The design must satisfy the following operating requirements and constraints:

- Drive system: A 1.5 HP, 110 VAC gear motor drives the shaft through a chain transmission.
- Sprocket arrangement: The drive sprocket is driven by a chain from the motor and engages a horizontal chain with a 180° angle of wrap.
- Maximum operating speed: 60 rpm at the mixing hook.
- Allowable shaft shear stress: 40 MPa.
- Allowable normal stress: 60 MPa.
- Construction: Stainless-steel components suitable for washdown environments, including the mixing bowl and newly designed hook.
- Drive shaft: Transmits power and torque from the chain drive to the mixing hook.
- Chain and sprockets: Transfer power and torque between the drive components.



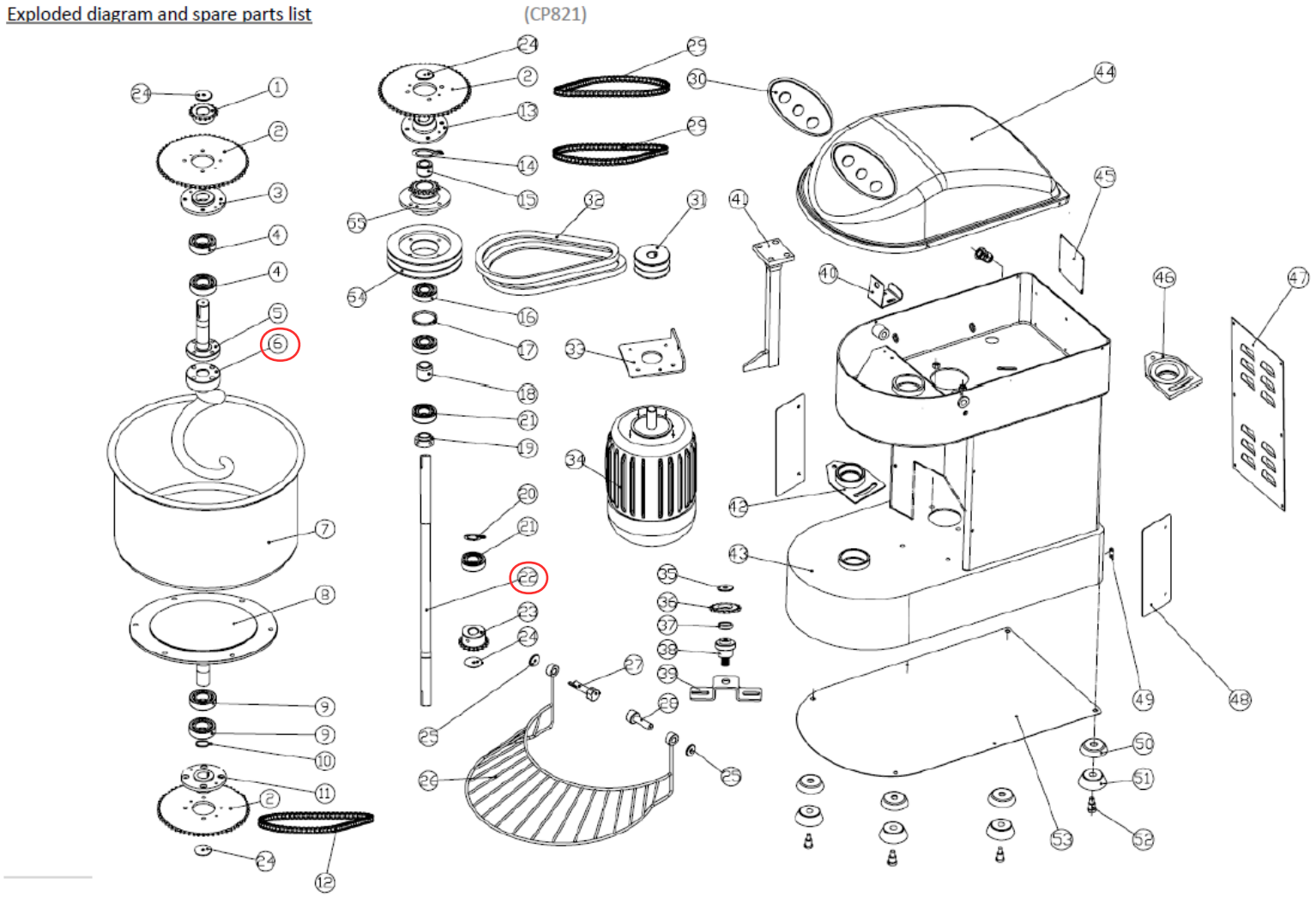




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Design Reference

Use the provided exploded schematic from the *Buffalo CP821* product listing as a reference to an existing commercial design. Study the arrangement and geometry of the main components, including bearing spacing, sprocket and gear locations, shaft diameters, and the connections between components. Use this reference to inform your design decisions without directly reproducing the existing assembly.



NO.	Description	QTY.
1.	Small chain wheel of main shaft	1
2.	Big chain wheel	3
3.	Spindle Sprocket Flange	4
4.	Bearing 6205-Z	2
5.	Main shaft	1
6.	Hook	1
7.	Bowl	1
8.	Bowl shaft	1
9.	Bearing 6205-Z	2
10.	Shaft ring 25	1
11.	Flanges of bowl shaft	1
12.	Chain 06B-1-92L	1
13.	Flanges of drive shaft	1
14.	Shaft ring 30	1
15.	Shaft Sleeve 2	1
16.	Bearing 6204-Z	2
17.	Bearing Spacer	1
18.	Shaft Sleeve 1	1
19.	Ring 20	1
20.	Shaft ring 20	1
21.	Bearing 1204-Z	2
22.	Drive shaft	1
23.	Small chain wheel of drive shaft	1
24.	Clamp plate 34	1
25.	Cover of fixed shaft	2
26.	Protect cover	1
27.	Left fixed shaft	1
28.	Right fixed shaft	1

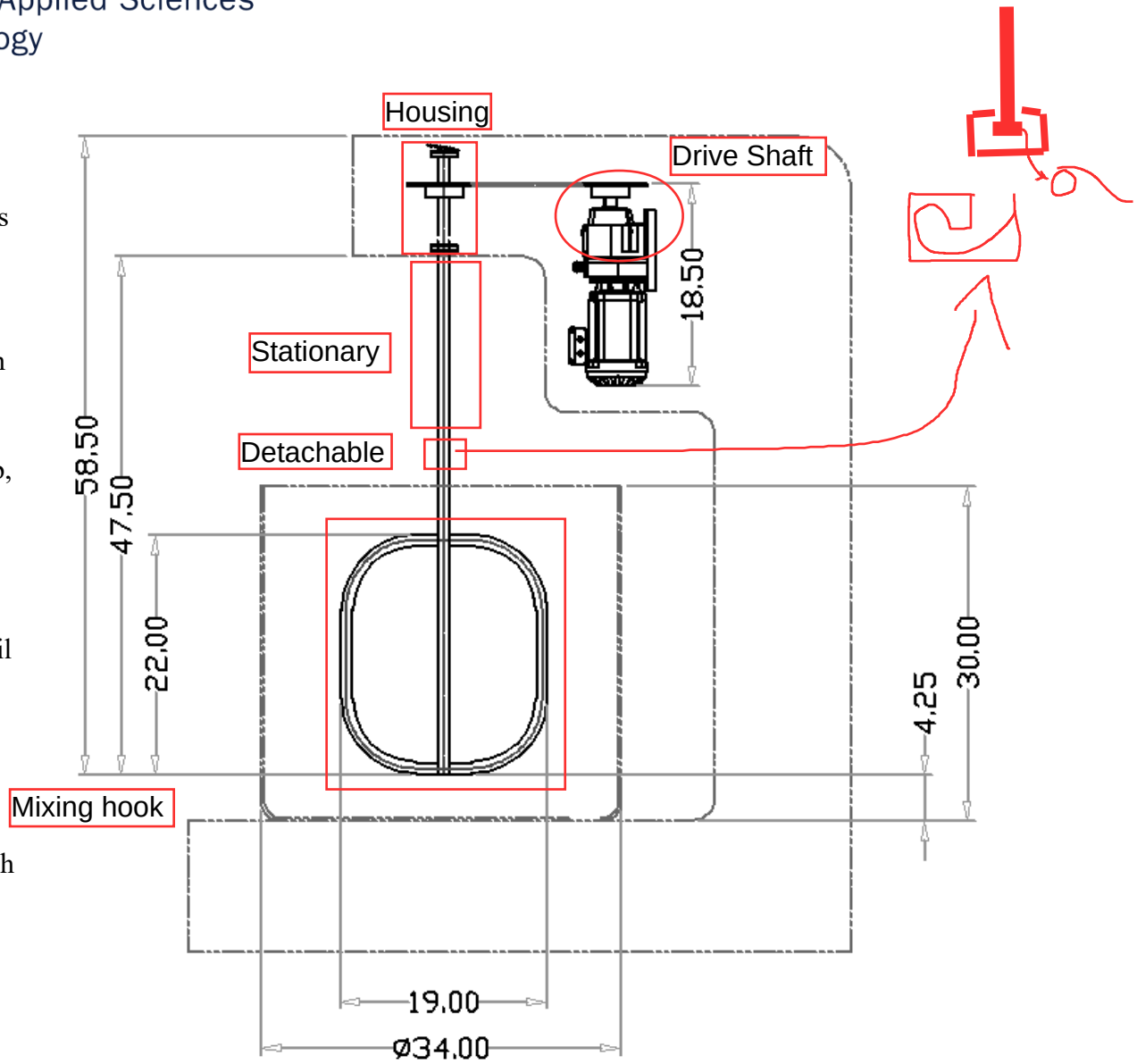
NO.	Description	QTY.
29.	Chain 06B-1-70L	2
30.	Control panel	1
31.	Motor belt wheel	1
32.	Belt A630	2
33.	Motor seat	1
34.	Motor	1
35.	Plate	1
36.	Chain wheel	1
37.	Chain wheel lining ring	1
38.	Chain wheel shaft	1
39.	Frame	1
40.	Switch seting board	1
41.	Fixed block	1
42.	Down bearing seat	1
43.	Housing	1
44.	Cover	1
45.	Brand	1
46.	Up bearing seat	1
47.	Back cover	1
48.	Lateral board	2
49.	Grounding mark	1
50.	Protect sets	4
51.	Anchor rubber pad	4
52.	Screw M8	4
53.	Bottom cover board	1
54.	Small chain wheel of drive shaft	1
55.	Drive pulley	1

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Objective

Using the provided schematic layout and dimensions as a reference, students are required to:

- Apply machine design principles to analyse and size the shafts, select suitable standard components, and develop a functional design that meets the specified performance and operating requirements.
- Using data and values from the previous step, design and simulate the shaft assembly, including the shaft housing and associated components.
- Prepare complete assembly and manufacturing drawings with sufficient detail to support purchasing, manufacturing, assembly, and inspection.
- Incorporate practical design features that allow the mixing hook to be easily installed and removed for frequent cleaning, and the mixing bowl to be removed at the end of each production batch for cleaning and maintenance.



Tasks / Deliverables

Submit a single PDF bundle containing all required documentation, along with an Excel calculation workbook or analysis sheet.

1. Calculation Workbook *(Excel spreadsheet recommended)*

   Include the following:

   - Shaft design: Determine the minimum practical shaft diameters and provide all supporting calculations.
   - Free-body diagrams (FBDs): Show the external forces, applied loads, and support reactions acting on the shafts.
   - Component sizing and analysis: Include the sizing and verification of keys, retaining rings, bearing seats, and other relevant components, along with any required simulation results.
   - Material selection: Justify the selected materials based on their mechanical properties, heat treatment, and suitability for the intended application.
   - Design assumptions and references: Document all assumptions, load cases, factors of safety, and references to the formulas, standards, and design methods used.

2. 2D Main Assembly Drawings

   - Main assembly drawing(s): Provide clear assembled views of the drive shaft, detachable mixing hook, associated components, and shaft housing within the relevant portion of the machine. Show the overall geometry, component locations, and relationships between components.
   - Detailed shaft drawings: Prepare two (2) separate, fully dimensioned engineering drawings:
     1. Lower detachable shaft section: Show the geometry and connection features required to attach and remove the mixing hook.
     2. Upper stationary shaft section: Show the geometry and features of the upper shaft section, including the relevant bearing seats and connections to associated components.

Submission Instructions

- Grading: This assignment is worth 10% of the overall course grade, with an additional 1% bonus mark available.
- Show all work required for each item and provide final conclusions in the designated areas. Use correct units, appropriate significant figures, and proper rounding.
- Marks may be deducted for incorrect notation, improper formatting, or failure to follow the specified requirements.
- Submit the complete assignment according to the following requirements:
    - Submission platform: Blackboard only. Submissions through other platforms or by email will not be accepted.
    - File format: Submit one complete file in PDF format.
- Submissions that do not comply with these requirements will not be accepted.

Marking module:

issue	Comment	Worth	marking
Using correct template Title block	○ To use Humber template, ○ Show correct scale ○ Complete title blocks		
Shaft calculation	○ Correct FBD (1.0 mark) ○ Correct torque and load diagrams (1.0 mark) ○ Correct Vertical and Horizontal BMD (1.0 mark) ○ Correct resultant BMD (0.5 mark) ○ Correct shaft sizing, and sound judgment. (1.0 mark)	4.5	
Shaft assembly 2D drawing.	○ Assembly drawing with All parts shown clearly (0.5 mark) ○ Implement correct components for a correct and functional design, (0.5 mark) ○ Simulate shaft housing clearly (0.5 mark) ○ Center lines missing / wrong. (max of -0.5 mark) ○ BOM and balloons to include part name, number, supplier and quantity (0.5 mark) ○ Not Utilizing the paper size efficiently, (max of -1.0 mark)	1.5	

	○ Wrong projections and sections, (max -1.5 marks) ○ Incorrect design (-2 marks) ○ insufficient dimensions for packaging and assembly preparation (-0.5 mark)		
Top Shaft drawing	○ Clear shaft design (1.0 mark) ○ specify material type, size and QTY (0.5 marks) ○ sufficient and correct dimensions for manufacturing (0.5 marks) ○ Not Utilizing the paper size efficiently, (max of -0.5 mark) ○ Wrong projections and sections, (max -1.5 marks) ○ Incorrect design (-1.5) marks	2.0	
Hook Shaft drawing	○ Clear shaft design (1.0 mark) ○ specify material type, size and QTY (0.5 marks) ○ sufficient and correct dimensions for manufacturing (0.5 marks) ○ Not Utilizing the paper size efficiently, (max of - 0.5 mark) ○ Wrong projections and sections, (max -1.5 marks) ○ Incorrect design (-1.5 marks)	2.0	
Quick drive shaft-hook assembly feature	○ Correct design (1 mark)	1.0	
		Overall %10 (max of 10 marks)	
NOTE: Will ignore and mark 0/10 for wrong file format submission			